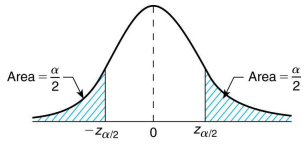
Agenda

- Earlier
  - Confidence intervals

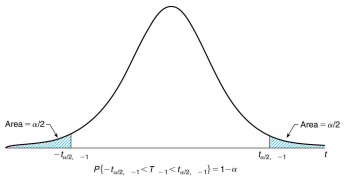
$$\Pr\left(-z_{0.025} \leq \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \leq z_{0.025}\right) = 0.95$$



1

Review: 95% confidence interval

- When we estimate a population parameter θ using an estimate $\hat{\theta}$, the defining property of a 95% confidence interval is
  - "a type of interval that contains the true value of θ for 95% of samples"
- In our examples: a confidence interval is the $\hat{\theta} \pm t$ standard errors
- This works exactly if the distribution is normal, and
  - "quite well" when the distribution is unknown
  - $(\bar{X}-\mu)/(S/\sqrt{n}) \sim t_{n-1}$
  - $\Pr\left(-t_{0.025,n-1} \leq \frac{\bar{X}-\mu}{S/\sqrt{n}} \leq t_{0.025,n-1}\right) = 0.95,$



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Review

TABLE 7.1 *100(1 −α) Percent Confidence Intervals*

$$X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$$

$$\overline{X} = \sum_{i=1}^{n} X_i/n, \qquad S = \sqrt{\sum_{i=1}^{n} (X_i - \overline{X})^2/(n-1)}$$

Assumption	Parameter	Confidence Interval	Lower Interval	Upper Interval
σ^2 known	μ	$\overline{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$	$\left(-\infty, \overline{X} + z_{\alpha}\frac{\sigma}{\sqrt{n}}\right)$	$\left(\overline{X} + z_{\alpha}\frac{\sigma}{\sqrt{n}}, \infty\right)$
σ^2 unknown	μ	$\overline{X} \pm t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}$	$\left(-\infty, \overline{X} + t_{\alpha,n-1}\frac{S}{\sqrt{n}}\right)$	$\left(\overline{X} - t_{\alpha,n-1}\frac{S}{\sqrt{n}}, \infty\right)$

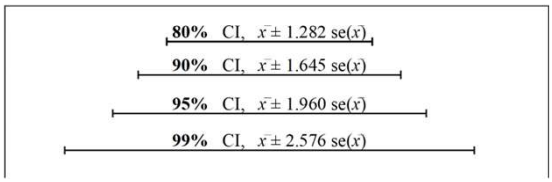
We use the empirical standard deviation in practice

Lower bound: at least as large

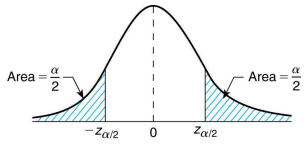
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Agenda

- Earlier
  - Confidence intervals
- Today
  - Comparing two populations or treatments



$$\Pr\left(-z_{0.025} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{0.025}\right) = 0.95$$



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Course details from IITB for CS215

- The role of statistics.
- Graphical and numerical methods for describing and summarising data.
- Probability.
- Population distributions.
- Sampling variability and sampling distributions.
- Estimation using a single sample.
- Hypothesis testing a single sample.
- *Comparing two populations or treatments.*
- Simple linear regression and correlation.
- Case studies.

Example: Comparing two populations

- Rheumatoid arthritis (RA) is a debilitating condition
- We measuring sulfhydryl concentration (mmol) in blood samples
- Statistical question:
  - What is the difference between concentration in those without and with RA?
  - Let's compute confidence intervals

	Without RA	With RA
	1.84	1.765
	1.92	3.015
	1.94	2.575
	1.92	2.225
	1.85	2.225
	1.91	2.715
	2.07	
Sample size	7	6
Sample mean	1.92143	2.42000
Sample standard deviation	0.07559	0.44049

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Example: Comparing two populations

- Statistical question: what is the difference between those without RA and those with rheumatoid arthritis?
  - n = 7, df = 6, t(0.025, 6) = 2.447 SE = 0.07559/√7 = 0.02857, margin = 2.447 × 0.02857 = 0.0699,CI = 1.92143 ± 0.0699 = (1.852, 1.991)
  - n = 6, df = 5, t(0.025, 5) = 2.571 SE = 0.44049/√6 = 0.17983, margin = 2.571 × 0.17983 = 0.4623 CI = 2.42 ± 0.4623 = (1.958, 2.882)
- The intervals overlap.
  - This might indicate that, for the without RA condition, the true mean is more than 1.958 (lower end of the with-RA condition)
  - This is not the way we compare two populations

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How different are the two population?

- Confidence interval for a difference between population means
  - $(\mu_1 - \mu_2)$: $(\bar{x}_1 - \bar{x}_2) \pm t\ \mathrm{se}(\bar{x}_1 - \bar{x}_2)$ ← Two means
- When two samples are independent the variance adds up, so

  $\mathrm{se}(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
- Conservative degree of freedom approximation for the t-distribution:
  - $\min(n_1 - 1, n_2 - 1)$ ← Employed in practice
- In the RA blood sample example:
  - Difference between means = 2.42 – 1.92143 = 0.49857, se = sqrt(0.07559²/7 + 0.44049²/6) = 0.1821, t(0.025, 5) = 2.571, margin=0.4681, CI = (0.030, 0.967) does not include zero, the with-RA always exceeds the without-RA

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Confidence intervals for difference

- Previous discussions assume empirical approximations
- If population is normal and means are estimated,
  - Assume: variances are known, then CI uses the known variances, they are added, and we work with exact normal distribution (see next slide)
  - Assume variances are unknown, and we use the empirical values
    - If they are different, then CI is complicated (o-o-s).
    - If they are same, then CI uses the pooled estimator and df = (n+m-2)
  - Note: Many cases variances are unknown and different!

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Review

TABLE 7.2 *100(1 − σ) Percent Confidence Intervals for $\mu_1 - \mu_2$*

$$X_1, \ldots, X_n \sim \mathcal{N}(\mu_1, \sigma_1^2)$$
$$Y_1, \ldots, Y_m \sim \mathcal{N}(\mu_2, \sigma_2^2)$$
$$\overline{X} = \sum_{i=1}^{n} X_i/n, \qquad S_1^2 = \sum_{i=1}^{n} (X_i - \overline{X})^2/(n-1)$$
$$\overline{Y} = \sum_{i=1}^{m} Y_i/n, \qquad S_2^2 = \sum_{i=1}^{m} (Y_i - \overline{Y})^2/(m-1)$$

Assumption	Confidence Interval
σ_1, σ_2 known	$\overline{X} - \overline{Y} \pm z_{\alpha/2}\sqrt{\sigma_1^2/n + \sigma_2^2/m}$
σ_1, σ_2 unknown but equal	$\overline{X} - \overline{Y} \pm t_{\alpha/2, n+m-2}\sqrt{\left(\frac{1}{n} + \frac{1}{m}\right)\frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}}$

Pooled estimator

Assumption	Lower Confidence Interval
σ_1, σ_2 known	$(-\infty, \overline{X} - \overline{Y} + z_{\alpha}\sqrt{\sigma_1^2/n + \sigma_2^2/m})$
σ_1, σ_2 unknown but equal	$\left(-\infty, \overline{X} - \overline{Y} + t_{\alpha, n+m-2}\sqrt{\left(\frac{1}{n} + \frac{1}{m}\right)\frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}}\right)$

Note: Upper confidence intervals for $\mu_1 - \mu_2$ are obtained from lower confidence intervals for $\mu_2 - \mu_1$.

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